DECO2850/DECO7285 Studio 2 - Interaction Design

Semester 2 2026

Technical Engineering and Execution Procedure

Smart Companion Bear - Electronics Build

Document No: SK-P1-003.1

Revision: Rev 2 (Week 11), realigned to the revised, evidence-based feature set agreed after Milestone 1.

Purpose: Working engineering reference for the Baobei smart bear. It turns the agreed feature set into a buildable wiring and assembly procedure, with the safety controls the build needs.

Status: Draft for team review. Component choices and pin assignments should be confirmed against the physical dev boards before soldering.

What changed in Rev 2

- **Features:** scope realigned to the kept features on the Baobei process map (v6). Camera/video is now formally ruled out, not left open.
- **Storage:** the bear now records to its own microSD card over SPI. The DFPlayer Mini is removed, because it can only play audio, not store recordings (Section 7).
- **Connectivity:** a hybrid model, with the bear's onboard Wi-Fi plus a cloud relay and local microSD store-and-forward (Section 7).
- **Power:** the bear now runs on three replaceable AA batteries. There is no charging inside the bear: the Li-po cell, charge module, magnetic dock and USB port are all removed. A buck-boost regulator keeps the supply steady as the batteries run down (Step 1).
- **Battery life:** the bear sleeps between uses and wakes when a paw is pressed or on a regular timer to check for messages (Step 8). The parent's app is warned when batteries run low.
- **Cooling:** no fan, and with no charging inside the bear there is no significant heat source left (Section 6).

- **Firmware behaviour rules:** added for the two formal commitments: child-initiated recording only, and local-first sync (Step 8).
- **Costing:** the BOM is updated, and the Rev 1 electronics subtotal is corrected (the line items summed to $119.25, not $118.44).

1. Scope and System Recap

The bear is the child's side of Baobei. It is a plush companion with no screen. The child squeezes a paw to record or play a voice message, and a heart LED glows when something new arrives. The bear keeps working when Wi-Fi drops and catches up when the connection returns. This procedure covers the physical build and the firmware *behaviour rules* the hardware must support. Detailed firmware code and the mobile app are covered separately (Section 11).

1.1 Feature-to-Hardware Traceability

Every component in this build exists to serve a kept feature. If a component cannot be traced to a row below, it should not be in the bear.

Kept feature	Status	Hardware that delivers it
Squeeze paws: record and play voice messages	Built	2 paw switches, INMP441 mic, MAX98357A amp, speaker
Heart LED: ambient response (new message, remote "light up")	Built	1 WS2812 NeoPixel
Power *(Rev 2: magnetic dock and charging removed)*	Changed	3 × AA batteries in a screw-covered holder, buck-boost regulator
Plush bear form: not a visible device	Built	Inner PLA enclosure, plush shell
Child-initiated recording only	Formalise	Mic powered/read only while the record paw is held (Step 8)
Local-first, offline-tolerant sync	Elevated: central to the redefined gap	microSD outbox/inbox + Wi-Fi + cloud relay (Section 7)
Voice notes and lullabies from the parent	App side, played on bear	Same playback path as above
Parent-only reply-status reminder	App side only (Milestone 2)	None on the bear, by design

1.2 Ruled Out of the Build

These features were removed after the Milestone 1 feature audit and must not be wired in:

- **Camera/video replies:** ruled out for bandwidth, data-safety and stigma reasons. Adding a camera later would be a separate hardware and firmware project, not a drop-in addition.
- **Hug detection, "warm" responses, mood tracking, games and streaks:** no evidence base, and warmth would need a heating element, which this build deliberately excludes.
- **Any reply-status display on the bear:** the child never sees "read", "delivered", "waiting" or "unanswered" states. The bear has no hardware for them, and the firmware must not repurpose the heart LED to show them.

1.3 Subsystems

Subsystem	Core components
Power	3 × AA batteries with on/off switch, 5V buck-boost regulator, switched peripheral rail, battery-voltage monitor
Core	ESP32-WROOM-32 dev board (FireBeetle ESP32)
Audio I/O	INMP441 I2S MEMS microphone, MAX98357A I2S class-D amplifier, 4Ω+ speaker
Sensors & LED	Two paw squeeze switches, one WS2812 (NeoPixel) LED
Connectivity & storage	Onboard 2.4 GHz Wi-Fi and BLE, microSD card on SPI (local store-and-forward)

2. Bill of Materials and Recommended Pin Map

Ref	Component	Notes
U1	ESP32-WROOM-32 dev board	34 usable GPIO; GPIO6–11 reserved for onboard SPI flash and must not be reused (Espressif Systems, n.d.-a)
U2	INMP441 I2S MEMS microphone breakout	1.8–3.3V logic, I2S digital output
U3	MAX98357A I2S class-D amplifier breakout	2.5–5.5V VIN, drives 4Ω+ speaker directly (Adafruit Industries, n.d.-a)
SPK1	Small speaker, ≥4Ω	Must not be lower impedance than 4Ω (Adafruit Industries, n.d.-a)
LED1	WS2812 / NeoPixel, single pixel	Chest "heart" indicator
SW1, SW2	Tactile push switches	Left paw: play. Right paw: hold to record, release to send
BAT1	3 × AA cells in an enclosed holder with on/off switch and screw cover	**Changed in Rev 2.** About 4.5V fresh, falling to about 3.0V when flat (alkaline), or 3.6V (rechargeable NiMH charged outside the bear)
U4	5V buck-boost regulator	**Changed in Rev 2.** Holds a steady 5V whether the batteries are above or below 5V. Replaces the 5V

Ref	Component	Notes
	(step-up/step-down)	boost converter
Q1	P-channel MOSFET load switch (or load-switch module)	**New in Rev 2.** Cuts power to the amp, LED, mic and microSD while the bear sleeps
R1, R2	100kΩ + 100kΩ divider	**New in Rev 2.** Battery-voltage monitor
—	~~Li-po cell, TP4056 charge module, magnetic dock~~	**Removed in Rev 2.** No charging inside the bear
U6	microSD breakout, 3.3V-native (SPI)	**New in Rev 2.** Local store-and-forward for recordings and received messages
—	~~DFPlayer Mini~~	**Removed in Rev 2.** Playback-only; cannot store recordings

Recommended pin assignments (confirm against your dev board's silkscreen before soldering; pin numbering varies slightly by board revision):

Signal	GPIO	Notes
I2S0 – Mic SCK (bit clock)	GPIO14	
I2S0 – Mic WS (word select)	GPIO15	
I2S0 – Mic SD (data out)	GPIO32	
Mic L/R select	GND	Selects left channel (Robogears, n.d.)
I2S1 – Amp BCLK	GPIO26	Second I2S peripheral, so mic and speaker can run independently
I2S1 – Amp LRC	GPIO25	
I2S1 – Amp DIN	GPIO22	
Amp SD (shutdown)	GPIO33	**New, optional.** Lets firmware switch the amp off when idle (less heat, less battery drain). Driving SD low shuts the amp down; driving it high (>1.4V) selects the left channel, so send mono audio on the left channel (Adafruit Industries, n.d.-a). Only wire this if your breakout exposes the SD pin; some modules don't
NeoPixel data	GPIO4	Through a 330–470Ω series resistor at the LED end (Adafruit Industries, n.d.-b)
Left paw switch (play)	GPIO13	Internal pull-up, switch to GND
Right paw switch (record)	GPIO27	Internal pull-up, switch to GND

Signal	GPIO	Notes
microSD SCK	GPIO18	VSPI default
microSD MISO	GPIO19	VSPI default
microSD MOSI	GPIO23	VSPI default
microSD CS	GPIO5	Strapping pin; the breakout's pull-up keeps it high at boot. Move to GPIO21 if boot problems appear
Battery voltage monitor (ADC)	GPIO34	Input-only pin, read through the 100k/100k divider (half the battery voltage)
Peripheral power enable	GPIO16	Drives Q1; switches the amp, LED, mic and microSD rail off during sleep

GPIO6–11 are wired internally to the module's SPI flash and are not available for general use (Espressif Systems, n.d.-b). GPIO34–39 are input-only and cannot drive outputs such as the LED or amplifier (Electronify India, n.d.). That makes GPIO34 a good fit for the battery monitor, which only needs to be read. GPIO16 (previously DFPlayer UART) now enables the peripheral rail; GPIO17 is free. The paw switches (GPIO13, GPIO27) are RTC-capable pins, so they can wake the ESP32 from deep sleep.

3. Pre-Build Safety Briefing

Read this section fully before any soldering or battery handling. Several of these hazards are specific to a battery-powered plush toy intended for a child aged 6–12, not just to the electronics bench.

3.1 Soldering Hazards

- Iron tips exceed 400°C and cause instant burns on contact with skin (Columbia University Research, n.d.). Never touch the tip or a freshly soldered joint — let joints cool before handling.
- Solder fumes (rosin/flux smoke) irritate eyes and airways; keep your face out of the smoke path and work in a ventilated area or with a fume extractor (Duke Pratt ECE Lab, n.d.; RPI EHS, n.d.).
- Wear ANSI-rated safety glasses — solder can spit (University of Illinois DRS, n.d.).
- Use a proper stand for the iron; never rest it on the bench, and unplug it whenever you step away (BE Labs, University of Pennsylvania, n.d.).
- If burned: cold running water for 15 minutes, no creams, cover with a clean dressing (Columbia University Research, n.d.).

3.2 Battery (AA) Hazards

- Use AA cells only. Never use button or coin cells anywhere in the bear. Swallowed button batteries cause severe internal injuries, and products containing them are covered by mandatory safety standards in Australia (ACCC, n.d.).
- Never mix old and new cells, different brands, or alkaline and NiMH cells in the same set. A weak cell can be driven into reverse and leak.
- Never try to recharge alkaline cells. If the team uses rechargeable NiMH cells, they are charged in a separate, off-the-shelf charger, never inside the bear.
- Use a holder with moulded polarity markings, and check polarity with the multimeter before every power-on.
- Remove the batteries if the bear will be stored for weeks. Old alkaline cells can leak corrosive fluid onto the electronics.
- Because this product is intended for a child, the batteries must sit behind a screw-fastened cover that the child cannot open by hand. Battery changes are an adult's job.

3.3 Toy-Safety and Child-Product Obligations

Because the artefact is a battery-powered electronic toy for a 6–12 year-old, the physical build should be evaluated against the small-parts and battery-containment expectations in the relevant toy-safety frameworks, even at prototype stage:

- Small internal components (switches, connectors, the LED, loose wiring) must not be reachable or removable by the child — this is the core small-parts/choking concern in toy safety standards for this age range (ASTM International, 2023; LISUN, n.d.).
- Battery compartments in commercial electric toys are increasingly required to resist a child prying them open — even at prototype stage, fasten your battery bay with real screws, not just tape or velcro (ACCC, n.d.; Federal Register, 2024).
- These are prototype-stage design disciplines, not a compliance claim — a Milestone 1/2 prototype does not need to pass formal certification, but building to these habits now avoids rework later and protects whoever handles it during testing.

3.4 Electrical and Moisture Risks

- Never solder on a live/powered circuit.
- Inspect the iron's cord before each session for fraying or damage (RPI EHS, n.d.).
- Because this is a soft-toy enclosure, condensation and handling moisture (little hands, drool, spilled drinks) are a real risk to exposed PCBs. Breakout boards should be conformal-coated or fully enclosed in a hard inner shell before the plush shell

goes over them — MEMS sensors themselves are typically sealed, but exposed traces and headers are not (Zbotic, 2026).

4. Tools and Workspace Setup

- Temperature-controlled soldering iron (lead-free solder: ~320–350°C tip temperature), stand, brass/wet sponge
- Fume extractor or well-ventilated bench
- Fine-tip tweezers, flush cutters, wire strippers
- Multimeter (continuity + voltage)
- Helping-hands or PCB vice
- Heat-shrink tubing + heat gun/lighter
- ANSI-rated safety glasses
- A fire-safe surface (ceramic tile) for the soldering area
- Anti-static mat/wrist strap (recommended, not mandatory for this parts list)

5. Step-by-Step Wiring and Assembly Procedure

Work in this order. Test each stage with the multimeter before applying power, and bench-test each subsystem on its own before final assembly into the plush shell.

Step 1 — Power Subsystem

1. Mount the 3 × AA holder (with its on/off switch) and solder its red lead to the buck-boost regulator's VIN and its black lead to GND. Check polarity twice.
2. Bench-test this stage alone first. Insert fresh batteries and confirm on the multimeter that the regulator outputs a steady 5V. Then test with a part-used set (or a bench supply at about 3.2V) and confirm it still outputs 5V. This is why a buck-boost is used: a plain boost converter can't regulate once fresh cells sit above 5V, and a plain step-down can't once they fall below it.
3. Take the regulator's 5V rail to the ESP32 board's VIN/5V pin. The ESP32 board's onboard regulator supplies 3.3V for logic.
4. Build the switched peripheral rail: the 5V rail feeds the MAX98357A VIN and NeoPixel VCC through Q1, and the ESP32's 3.3V feeds the INMP441 and microSD breakout through a second channel of the load switch (or a second MOSFET). Q1 is driven by GPIO16. With GPIO16 off, these parts draw no power while the bear sleeps.

5. Build the battery monitor: battery + → 100kΩ → GPIO34 → 100kΩ → GND. The pin reads half the battery voltage, which stays within the ESP32's ADC range. The divider draws about 25µA, small enough to leave connected.

Step 2 — ESP32 Core Bring-Up

1. Before wiring anything else, flash a minimal blink/serial-print sketch to the bare ESP32 board over USB and confirm it boots. This separates core failures from wiring failures later.
2. Mark GPIO6–11 as off-limits in your wiring notes (SPI flash) and GPIO34–39 as input-only (Espressif Systems, n.d.-a; Electronify India, n.d.).

Step 3 — Audio Output (MAX98357A + Speaker)

1. Solder BCLK, LRC, DIN, GND and VIN header pins onto the MAX98357A breakout if they are not pre-populated.
2. Wire BCLK→GPIO26, LRC→GPIO25, DIN→GPIO22, GND→common ground, VIN→5V rail. If the breakout exposes SD, wire SD→GPIO33.
3. Leave the GAIN pin unconnected for the default 9dB gain (Adafruit Industries, n.d.-a). Cap the maximum volume in firmware as well. A companion toy is held close to a child's ear, and lower volume also means less heat.
4. Solder the speaker leads to the amplifier's +/- output terminals. Do not connect this output to any other amplifier stage; it drives the speaker directly (Adafruit Industries, n.d.-a).
5. Flash a basic I2S tone-generation sketch and confirm audio output before moving on.

Step 4 — Audio Input (INMP441 Microphone)

1. Wire VDD→3.3V, GND→common ground, L/R→GND (selects left channel) (Robogears, n.d.).
2. Wire SCK→GPIO14, WS→GPIO15, SD→GPIO32.
3. Flash a basic I2S read-and-print sketch, speak near the mic, and confirm non-zero sample values in the serial output.

Step 5 — Sensors (Paw Switches and Battery Monitor)

1. Wire one leg of each tactile switch to GND and the other to its GPIO (GPIO13 left, GPIO27 right), using the ESP32's internal pull-up resistors in firmware. No external resistor is needed.
2. Test each switch with a simple digital-read sketch before embedding it in the paw.
3. Flash a sketch that reads GPIO34 and prints the battery voltage (multiply the pin reading by 2). Compare it with a multimeter reading across the battery holder and note the calibration offset. The ESP32's ADC is not precise, so calibrate once per board.

Step 6 — LED (NeoPixel Heart)

1. Solder a 330–470Ω resistor in series on the data line, as close to the LED as physically possible (Adafruit Industries, n.d.-b; ComponentIndex, 2026).
2. Wire VCC→5V rail, GND→common ground, DIN→GPIO4 (through the resistor).
3. If you later chain more than one pixel, add a 1000µF capacitor across the 5V/GND rail at the LED end to absorb inrush current (Adafruit Industries, n.d.-b).
4. Run the Adafruit NeoPixel library's basic colour-cycle example to confirm.

Step 7 — Storage (microSD on SPI)

Rev 2 replaces the DFPlayer path. The DFPlayer can play files from its own card, but the ESP32 cannot write recordings to that card. Local-first sync needs the bear to save recordings itself, so the ESP32 must own the SD card.

1. Wire the microSD breakout: VCC→3.3V, GND→common ground, SCK→GPIO18, MISO→GPIO19, MOSI→GPIO23, CS→GPIO5. Use a 3.3V-native breakout; 5V-only modules with level shifters behave unpredictably on a 3.3V supply.
2. Format a microSD card as FAT32 and create three folders: `/outbox` (child recordings waiting to upload), `/inbox` (parent messages waiting to be played) and `/played` (messages kept briefly for replay).
3. Flash the ESP32 Arduino core's SD example sketch. Confirm that the bear can write, read back and delete a test file.
4. Play a WAV file from `/inbox` through the MAX98357A to confirm the full local playback path, with no DFPlayer and no network.

Step 8 — Firmware Behaviour Rules (Kept Features)

These are not code; they are the rules the firmware must follow. They turn the two **[formalise]** commitments into something testable.

Child-initiated recording only

- The microphone is read only while the right paw is physically held down. Releasing the paw ends the recording and queues it in `/outbox`.
- There is no command, app message or cloud call that can start the microphone remotely. This is enforced by the absence of that code path, not by a setting someone could change.
- The heart LED shows a distinct "listening" colour while recording, so the child always knows when the bear is listening.

Local-first, offline-tolerant sync (store-and-forward)

- Every recording is saved to `/outbox` first, and only then uploaded. Upload failure never loses a message; the file stays in `/outbox` and retries when Wi-Fi returns.
- Parent messages are downloaded to `/inbox` when online and play from the card, so playback never depends on a live connection.

- Files are deleted from the card once delivery is confirmed (outbox) or after a short replay window (played), to minimise data held on the device.

Heart LED behaviour

- The LED uses separate patterns for: "listening" (recording), "sent" (a brief acknowledgement after release), "new message" (a gentle glow until played) and "light up" (the parent's remote trigger).
- The bear has no "waiting", "read" or "unanswered" state. **[team decision]** The heart should not light *only* for replies; otherwise a dark heart after several days becomes a "no reply" signal. An occasional bedtime glow or the remote "light up" trigger helps keep the heart from becoming a reply indicator.

Power (battery life)

Always-on Wi-Fi would flatten three AA cells in a day or two, so the bear spends most of its time in deep sleep:

- **Wake on paw:** pressing either paw wakes the bear immediately (both paw pins are RTC-capable). After a recording, the bear tries to upload it straight away, then goes back to sleep. If Wi-Fi is down, the file waits in `/outbox` for the next wake.
- **Wake on timer:** the bear wakes every 15 minutes **[team decision]** to check for parent messages, download them to `/inbox`, and upload anything waiting. Parent messages and the "light up" trigger therefore arrive at the next check-in, not instantly. That suits Baobei's asynchronous, no-pressure design, but the team should agree on the interval.
- **Peripherals off while asleep:** GPIO16 switches off the amp, LED, mic and microSD rail before sleep.
- **"New message" glow:** shown as a slow, brief pulse every few seconds rather than a constant light, and only for a set period (for example, until bedtime), to save power.
- **Low battery:** when the monitored voltage drops below the threshold found in Step 9, the bear sends a "batteries low" note to the **parent's app**. The child sees nothing, so the bear never signals a problem the child can't fix.

Step 9 — Full Bench Integration Test

1. With all subsystems on a shared ground and the shared 5V/3.3V rails, power up from the AA batteries only (not USB) and confirm:
   - Holding the right paw records (the "listening" LED shows), and releasing it saves the file to `/outbox` and shows "sent".
   - With Wi-Fi switched off, recordings still save, and they upload automatically when Wi-Fi returns.
   - A parent message arriving from the app lands in `/inbox`, the heart glows, and pressing the left paw plays it.
   - The app's "light up" trigger lights the heart.
   - Nothing on the bear can start the microphone except the paw.

2. Measure current at the battery in three states with the multimeter: deep sleep, a timer check-in (Wi-Fi on), and recording or playback. Use these numbers to estimate battery life (Section 6.2) and to set the low-battery threshold.
3. Only once this bench test passes, proceed to the warmth and battery-life check.

Step 10 — Warmth and Battery-Life Check (New in Rev 2)

1. Assemble the electronics inside the inner PLA enclosure and close the plush shell (or wrap it in the same stuffing).
2. Run the busiest case for 30 minutes: repeated recording and playback with Wi-Fi active. Touch-check the outer fabric; it should feel no warmer than skin temperature.
3. Then leave the bear running normally (sleeping, with timer check-ins) on a fresh set of batteries and log the battery voltage each day until the low-battery alert fires. This gives the team a measured battery life to report, instead of an estimate.

Step 11 — Enclosure Integration into the Plush Shell

1. Conformal-coat or box each bare breakout board before it goes near fabric or stuffing (Zbotic, 2026).
2. Mount the AA holder in a screw-fastened compartment the child cannot open by hand (ACCC, n.d.; Federal Register, 2024). Place it at the rear seam behind a fabric flap with a snap or hook-and-loop strip, so an adult can change batteries without unstitching the bear.
3. Put the on/off switch inside the same compartment, so the bear can be fully switched off for storage or travel. No wiring should be exposed at the plush surface.
4. Make the microSD card reachable only by an adult (inside the screwed compartment). It holds voice recordings.
5. Secure all connectors with hot glue or heat-shrink strain relief before final stuffing. A loose header pin becomes a small-parts hazard once sewn inside a toy (ASTM International, 2023).
6. Re-run the Step 9 integration test after the shell is closed, since Wi-Fi antenna performance and speaker resonance can both change once the electronics are inside fabric and stuffing.

6. Thermal Management and Battery Life

6.1 Is a Fan Needed?

No. With charging removed from the bear, the largest heat source in the earlier design is gone. What remains is small and brief:

Source	When	Heat
ESP32	Wi-Fi bursts at each check-in	Short peaks; the board sleeps the rest of the time

Source	When	Heat
MAX98357A amplifier and speaker	Only during playback	Low at the capped volume a companion toy needs
Buck-boost regulator	Whenever the bear is awake	Small conversion losses
NeoPixel (single pixel)	Brief pulses	Negligible at a soft glow

A fan would also cause new problems:

- **Noise:** the microphone sits millimetres away. Fan noise would end up in every recording, and a humming bear undermines a calm companion at bedtime.
- **Child safety:** a fan is a moving part with an air intake inside a toy, which means hair, fibres and small fingers.
- **Reliability:** polyester stuffing fibres clog fan intakes quickly.
- **Battery life:** a fan would drain the AA cells faster than everything else combined.

The vented PLA inner enclosure and the warmth check in Step 10 are enough.

6.2 Battery Life

Battery life is now the main power constraint, replacing heat. The figures below are rough planning numbers only; replace them with the measurements from Steps 9 and 10.

Mode	What drains the batteries	Rough battery life
Wi-Fi always on	The ESP32 radio, all day	About 1–2 days. Not acceptable
Deep sleep, 15-minute check-ins, ~10 uses a day	Short Wi-Fi bursts, plus recording and playback	Roughly 1–3 weeks on alkaline AAs

Choosing cells: alkaline AAs are fine for the prototype and exhibition. For a family using the bear every day, rechargeable NiMH AAs charged in an ordinary external charger cost less over time and create less waste, while still keeping all charging outside the bear. **[team decision]**

7. Connectivity and Storage Decision

Recommendation: a hybrid model. Wi-Fi plus a cloud relay carries messages between bear and app; a microSD card on the bear stores messages locally so nothing depends on a live connection. This is the only option that delivers both kept features: messages reach a parent in another city, *and* the bear keeps working offline.

7.1 Options Compared

Option	Reaches a parent far away?	Works offline (local-first)?	Fits the ESP32 and toy?	Verdict

Option	Reaches a parent far away?	Works offline (local-first)?	Fits the ESP32 and toy?	Verdict
SSD (solid-state drive)	No: it is local storage only	Yes	No: the ESP32 has no SATA/NVMe interface, and an SSD is far too large and power-hungry for a toy	Not feasible
microSD only	No: the messages never leave the bear	Yes	Yes	Fails the core use case
Cloud only (stream directly, no local storage)	Yes	No: messages are lost or stuck when Wi-Fi drops	Yes	Breaks the local-first commitment, which is central to the redefined gap
Hybrid: microSD + Wi-Fi + cloud relay	Yes	Yes	Yes: ~$6 of extra hardware and the Wi-Fi already on the chip	**Recommended**

If "SDD" in team discussions meant an SD card rather than a solid-state drive, the microSD in this design is that card.

7.2 How It Works

1. **Child records:** the recording is saved to `/outbox` on the microSD card.
2. **Bear uploads when online:** the file goes over Wi-Fi to the cloud relay (Firebase Storage for the audio, Firestore for the message record, matching the app stack in Section 11). If Wi-Fi is down, the file waits in `/outbox`.
3. **Parent is notified:** the app receives a push notification and plays the message.
4. **Parent replies:** the reply uploads to the cloud. The bear downloads it to `/inbox` at its next sync, and the heart glows.
5. **Cleanup:** files are deleted from the bear once delivered or replayed, and cloud-side retention is kept short.

7.3 Sizing and Bandwidth

- **Audio format:** 16 kHz, 16-bit mono WAV is about 32 KB per second, so a 30-second message is roughly 1 MB. That is fine on home Wi-Fi.
- **Storage:** a 16GB card holds well over 100 hours of audio at that rate, far more than store-and-forward needs. Card size is not a constraint.

- **Low-bandwidth households:** if uploads are slow, encode to IMA-ADPCM (about 4:1 compression) on the bear. **[team decision]** Only needed if field testing shows slow uploads.

7.4 Wi-Fi in Practice

- **No extra hardware:** the ESP32's onboard 2.4 GHz Wi-Fi and BLE are enough. Setup uses the BLE pairing flow in Section 10.
- **2.4 GHz only:** the ESP32 cannot join 5 GHz-only networks. Most home routers broadcast both.
- **For the exhibition and demos:** university enterprise networks (such as eduroam) use WPA2-Enterprise logins, which are awkward on an ESP32. Use a phone hotspot for demos.
- **Households without Wi-Fi:** a caregiver's phone hotspot works for periodic syncs, and local-first storage means the bear is still usable between syncs. A cellular module would remove the Wi-Fi dependency, but it adds cost, power draw and a SIM plan, so it is out of scope for this prototype.

7.5 Cloud Service Note

The app stack in Section 11 uses Firebase. Cloud Storage for Firebase now requires the pay-as-you-go Blaze plan for new projects, though it includes a no-cost usage allowance that a student prototype should stay within (Firebase, n.d.). Set a budget alert of $1 when the project is created. Supabase, which the earlier web prototypes used, is a workable alternative. **[team decision]** Pick one with the app subteam so the bear and app share the same backend.

8. Risk Register

Risk	Likelihood	Severity	Mitigation
Batteries run flat too quickly *(new)*	Medium	High (bear goes silent)	Deep sleep with paw and timer wake, peripherals switched off while asleep, measured battery life (Steps 9–10), low-battery alert to the parent's app
Battery leakage or corrosion *(new)*	Low	Medium	Never mix old/new or alkaline/NiMH cells; remove cells for long storage; holder inside the inner enclosure, away from boards
Parent messages and "light up" delayed by sleep *(new)*	Medium	Low–Medium	Agreed check-in interval; paw press triggers an immediate sync; the delay fits the asynchronous design
Solder burns	Medium	Medium	Iron-stand discipline, cool-down time, first-aid steps documented above

Risk	Likelihood	Severity	Mitigation
Flux fume exposure	Medium	Low–Medium	Ventilated bench or fume extractor; avoid leaning over the joint (RPI EHS, n.d.)
Loose small parts (switches, header pins, LED, microSD card) becoming a choking hazard	Medium	High (child product)	Heat-shrink/hot-glue strain relief, screw-fastened compartments, microSD card inside the adult-only compartment (ASTM International, 2023)
Battery compartment opened by a child	Low–Medium	High	Screw-fastened AA compartment behind the rear flap; battery changes by adults only; AA cells only, never button or coin cells (ACCC, n.d.; Federal Register, 2024)
Moisture ingress (handling, drool, spills)	Medium	Medium	Conformal coating or a hard inner enclosure for all boards (Zbotic, 2026)
Batteries inserted the wrong way round	Low	Medium	Holder with moulded polarity markings; multimeter check before first power-on; adult-only battery changes
GPIO conflict with SPI flash pins (bricking the board)	Low, if pin map followed	Medium	Avoid GPIO6–11 entirely; confirm against the board silkscreen before wiring (Espressif Systems, n.d.-a)
Boot failure from the microSD CS on strapping pin GPIO5 *(new)*	Low	Low	Breakout pull-up keeps it high at boot; move CS to GPIO21 if boot problems appear
Wi-Fi timing interference with the NeoPixel data line	Low	Low	Use the RMT or I2S driver for LED output rather than bit-banging (ComponentIndex, 2026)
Messages lost when Wi-Fi drops *(new)*	Medium (rural households)	High (core claim)	Store-and-forward: save to microSD first, upload second, delete only after confirmed delivery (Step 8)
Microphone triggered without the child's action *(new)*	Low	High (trust and privacy)	No remote code path to start the mic; "listening" LED while recording; tested in Step 9
Voice recordings exposed if the bear or card is lost *(new)*	Low	Medium (privacy)	Delete files after delivery, card inside the screwed compartment, short cloud retention
Unauthenticated pairing / Wi-Fi	Low	High (privacy)	Per-device pairing token generated at first pair (not a fixed default);

Risk	Likelihood	Severity	Mitigation
credential exposure (a stranger's app claims or eavesdrops on a bear over shared Wi-Fi)			cloud security rules scoped to the matched bear_id/parent_account_id pair (Espressif Systems, n.d.-c)

9. Costed Bill of Materials (AUD)

Pricing is per single bear unit, checked against Australian retailers at the time of writing. Prices marked "(estimate)" are typical market rates rather than a specific live listing. Core Electronics prices are GST-inclusive. Shipping is not included; a single consolidated order typically adds $7–11.

9.1 Electronics

	Component	Price (AUD)	Source
	FireBeetle ESP32 dev board (DFR0478)	$18.70	https://core-electronics.com.au/firebeetle-esp32-iot-microcontroller-supports-wi-fi-bluetooth.html
	INMP441 I2S MEMS microphone (CE10259)	$12.30	https://core-electronics.com.au/i2s-microphone-inmp441.html
	MAX98357 I2S amplifier module (DFR0954)	$10.15	https://core-electronics.com.au/max98357-i2s-amplifier-module.html
	4Ω+ speaker, small	$8.95	https://core-electronics.com.au/mono-enclosed-speaker-3w-4-ohm.html
	3 × AA holder with on/off switch and screw cover *(new)*	~$4.00 (estimate)	Source to confirm
	3 × AA alkaline batteries *(new; recurring)*	~$4.00 (estimate)	Any supermarket
	5V buck-boost regulator *(new)*	~$15.00 (estimate)	Source to confirm
	P-channel MOSFET load	~$2.00 (estimate)	Source to confirm

	Component	Price (AUD)	Source
	switch + resistors *(new)*		
	microSD card, 16GB	~$12.00 (estimate)	https://core-electronics.com.au/sd-microsd-memory-card-16gb-class-10.html
	microSD breakout, 3.3V SPI *(new)*	~$6.00 (estimate)	Source to confirm
	~~Magnetic pogo-pin charging dock (2-pin)~~ *(removed)*	~~$8.00~~	—
	NeoPixel, single pixel	~$3.50 (estimate)	https://core-electronics.com.au/ultra-bright-3-watt-chainable-neopixel-led-ws2811.html
	Paw tactile switches × 2	~$2.00 (estimate)	https://core-electronics.com.au/soft-tactile-button-8mm-x-10.html
	Wire, resistors, heat-shrink, hot glue allowance	~$10.00 (estimate)	https://core-electronics.com.au/added/?product=419
	~~Li-po battery, TP4056 charge module, 5V boost converter~~ *(removed)*	~~$25.10~~	—
	~~DFPlayer Mini (DFR0299)~~ *(removed)*	~~$8.55~~	—
	Electronics subtotal	**≈ $108.60**	

9.2 Physical Shell Materials

Material	Price (AUD)	Source
Faux fur fabric, short/medium pile, ~0.5m	~$12–15 (current Lincraft sale rate is $12/m)	Lincraft
Polyester fibre toy stuffing, 500g	~$10.00 (estimate)	Lincraft
PLA filament for the vented internal enclosure, ~150g of a 1kg spool	~$5.50	Core Electronics
Felt, thread, safety eyes, trim allowance	~$8.00 (estimate)	—

Material	Price (AUD)	Source
Fasteners/screws for the battery compartment	~$5.00 (estimate)	—
Shell materials subtotal	**≈ $40.50**	

9.3 Grand Total

≈ $149 AUD per bear, before shipping, down from Rev 1's ≈ $159. The DFPlayer, magnetic dock, Li-po battery, charge module and boost converter are removed; the microSD breakout, AA holder, buck-boost regulator and load switch are added; and the Rev 1 subtotal has been corrected. Batteries are a running cost for families (about $4 per set, or a one-off cost for rechargeable NiMH cells and a charger). Cloud costs should stay at $0 within the no-cost allowance at prototype scale (Section 7.5).

Things to double-check before committing this to the shared repo:

- The buck-boost regulator and AA holder prices are estimates; confirm them with a supplier.
- Check that the AA holder fits the plush cavity alongside the inner enclosure. Three AA cells are bulkier than the flat Li-po they replace.

10. Device Pairing and App Connectivity

A teddy bear has no keyboard or screen, so it can't be handed Wi-Fi credentials directly. The parent's app also needs a reliable way to know which physical bear it's talking to. Both problems are solved by a short-range pairing step that happens once, before the bear ever talks to the cloud. **[team decision]** This security model has not yet been reviewed by the whole team.

10.1 Two-Layer Connection Model

No new hardware is required. The ESP32 already has both Wi-Fi and Bluetooth Low Energy (BLE) radios on the same chip (Espressif Systems, n.d.-a).

Pairing/setup layer: BLE. Used only during first-time setup (and again after a factory reset). The bear advertises itself, and the app connects locally over Bluetooth to hand over Wi-Fi credentials and a pairing token. No internet connection is needed for this step.

Operational layer: Wi-Fi + cloud relay. Once paired, all voice-message sync runs over Wi-Fi to the cloud relay, with microSD store-and-forward when offline (Section 7). BLE can idle to save power.

10.2 Pairing Sequence

1. **App: open pairing mode.** The parent taps "Add bear" in the app.
2. **Bear: BLE advertising.** The bear broadcasts a unique bear ID over BLE.

3. **App: send Wi-Fi + token.** The app connects over BLE and writes the home Wi-Fi SSID, password and a freshly generated pairing token to the bear.
4. **Bear: join Wi-Fi and register.** The bear saves the credentials to flash (NVS), connects to Wi-Fi, and calls the cloud relay with its bear ID and pairing token.
5. **Cloud: confirm pairing.** The relay links bear_id to parent_account_id and confirms back to both devices.
6. **App: bear connected.** Ongoing sync now runs over Wi-Fi and the cloud relay.

This is the standard pattern implemented by Espressif's wifi_provisioning component in ESP-IDF (Espressif Systems, n.d.-c), so it does not need to be built from scratch.

10.3 Security Considerations

- The pairing token must be generated fresh per device at first pair, not shipped as a fixed default. A hardcoded token would let any app claim any bear.
- Cloud security rules must scope every message read and write to the matched bear_id/parent_account_id pair, so a bear can't be read or written to by an account it isn't paired with (Espressif Systems, n.d.-c).

11. Mobile App Architecture

Status: proposed stack, not yet confirmed with the team. This section covers the app side of the same pairing and sync relationship.

11.1 Recommended Stack

Layer	Choice	Why
Framework	Flutter (Dart)	Single codebase for iOS and Android, with a hot-reload workflow suited to an iterative studio timeline (Flutter, n.d.)
BLE	flutter_blue_plus	Maintained plugin for scanning, connecting and reading/writing BLE characteristics; drives the Section 10 pairing flow (pub.dev, n.d.-a)
Auth	firebase_auth	Parent account sign-in
Message data + sync	cloud_firestore	Cloud database with live sync and built-in offline persistence (pub.dev, n.d.-b)
Audio storage	firebase_storage	Object storage for voice-message audio (pub.dev, n.d.-c)
Push notifications	firebase_messaging	Notifies the parent when a new message arrives from the bear (pub.dev, n.d.-d)
Secure local storage	flutter_secure_storage	Stores the pairing token in the OS keychain/keystore (pub.dev, n.d.-e)

Alternative: React Native is a reasonable substitute if the team has more JavaScript than Dart experience, swapping flutter_blue_plus for react-native-ble-plx and the FlutterFire

plugins for the React Native Firebase SDK. Choose based on the team's comfort, not a perceived "better" option.

11.2 Software Architecture (Four Layers)

- **Connectivity layer:** talks to the bear over BLE during pairing only, and to the cloud over the Firebase SDKs after that. This is the only layer that knows about either external system.
- **State and sync manager:** holds app-wide state (current bear, pairing status, message list) and the offline send queue, and decides what the UI shows.
- **Presentation layer:** screens and UI state only, with no direct network or BLE calls.
- **Local storage:** a message cache (for offline viewing, non-sensitive) and a secure token store (the pairing token, sensitive). These are kept separate.

11.3 Screen Map (Aligned to the Kept Features)

Screen	Purpose	Kept feature
Onboarding and bear pairing	Runs the BLE pairing sequence from Section 10.2	Supports all features
Message timeline	Sent and received voice messages, newest first, with a delivery status for the parent	Message status (confirms delivery)
Record and send	Record a voice note or lullaby, with a playback preview before sending	Send a voice note (incl. lullaby)
Light up the bear	One tap to make the bear's heart glow at its next check-in	Light up the bear (remote LED trigger)
Reply-status reminder	A gentle, parent-only prompt when a child's message hasn't been answered. Never shown on the bear	Reply-status indicator (Milestone 2 build target)
Settings	Account, notifications, low-battery alerts and last check-in time (parent-only diagnostics)	—

Features ruled out in the Milestone 1 audit (mood widgets, streaks, coins, games, hug detection, "warm" responses) must not be added to the app.

11.4 Offline and Sync Behaviour

Cloud Firestore's offline persistence caches recently synced documents, so the message timeline still renders without a connection; outgoing messages queue locally and send automatically once connectivity returns (pub.dev, n.d.-b). This mirrors the bear's microSD store-and-forward (Section 7), so offline tolerance is consistent end to end.

11.5 Security

- The pairing token is written to flutter_secure_storage (OS keychain on iOS, encrypted keystore on Android), never to plain preferences storage (pub.dev, n.d.-e).

- Parent accounts authenticate through Firebase Auth. Firestore and Storage security rules are scoped per parent_account_id, matching the access-control model in Section 10.3.

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